UNIT 4: ER MODEL

Question Bank with Answers and Worked Conversions

Easy • Medium • Hard • ER-to-Relational Mapping • Relational-Schema-to-ER Reasoning • Tutorial Solutions

Primary reference: Unit4_ERmodel_final.pdf (103 pages)

Prepared for university DBMS examination practice

How to Use This Question Bank

This question bank follows the terminology, examples, and mapping rules in the supplied Unit 4 ER Model slides. Attempt each question before reading its answer. For conversion questions, first identify entity sets and keys, then relationships, cardinalities, and participation; only then apply the relational mapping rules.

Source coverage note: The uploaded reference has 103 pages. Its final text section contains five explicit tutorial questions, and the slides also include labeled conversion exercises and several diagram-based examples. The final tutorial section below answers the five explicitly listed questions and works through the conversion examples whose schema is legible in the slides. Some slides titled “Exercise—ER Model for Banking/University/Company Database” contain diagrams with little or no extractable text; where the image does not expose a complete written question or legible diagram detail, this bank does not pretend to reconstruct hidden requirements. Use the page references provided to cross-check the original.

Notation used: [PK] = primary key; [FK] = foreign key; underlined attributes represent primary keys in the ER/schema convention. A double line means total participation. A double diamond denotes an identifying relationship for a weak entity. A dashed underline marks a weak entity’s partial key.

Exam method

• Mark every entity noun and its descriptive attributes in the scenario.

• Choose a stable identifier; distinguish super keys from minimal candidate keys.

• Read cardinality in both directions; do not confuse “one-to-many” with participation.

• Decide whether participation is total (minimum 1) or partial (minimum 0).

• Map strong entities first, then 1:N and 1:1 relationships, M:N relationships, multivalued attributes, and weak entities.

• State assumptions whenever the wording does not specify a constraint.

Easy Level — MCQs

Q1. Which statement best describes an entity?

A collection of columns only

A distinguishable real-world object

A relationship between tables

A permitted value range

Correct answer: B. An entity is a real-world object distinguishable from other objects.

Q2. A set of all students with the same descriptive properties is a:

Domain

Entity set

Relationship attribute

Candidate key

Correct answer: B. An entity set groups entities of the same type.

Q3. The set of permitted values for an attribute is its:

Role

Domain

Degree

Discriminator

Correct answer: B. A domain defines the allowed values of an attribute.

Q4. Which ER symbol represents an entity set in the slide notation?

Diamond

Oval

Rectangle

Double line

Correct answer: C. Rectangles represent entity sets.

Q5. Which ER symbol represents a relationship set?

Diamond

Rectangle

Dashed oval

Arrow only

Correct answer: A. A diamond represents a relationship set.

Q6. An underlined attribute in the ER diagram denotes the:

Foreign key

Derived attribute

Primary key

Multivalued attribute

Correct answer: C. The slides use underlining to mark primary-key attributes.

Q7. Which is a simple attribute?

Full name with first and last name

Address with street and city

State

Name split into components

Correct answer: C. A simple attribute is atomic or indivisible in the model.

Q8. Which is a composite attribute?

Salary

Student ID

Address with street, city, state and zip

Age calculated from date of birth

Correct answer: C. A composite attribute has meaningful component attributes.

Q9. Which attribute can contain several values for one entity?

Single-valued

Multivalued

Derived

Primary

Correct answer: B. A multivalued attribute stores a set of values, such as phone numbers.

Q10. Age calculated from date of birth is:

Composite

Multivalued

Derived

A relationship

Correct answer: C. A derived attribute is computed from other stored attributes.

Q11. The association between an instructor and a student is a:

Domain

Relationship

Candidate key

Weak entity

Correct answer: B. A relationship represents an association among entities.

Q12. A relationship involving two entity sets has degree:

1

2 / binary

3 / ternary

n-ary only

Correct answer: B. A binary relationship involves two entity sets.

Q13. A relationship involving three entity sets is:

Unary

Binary

Ternary

Weak

Correct answer: C. Degree 3 is called ternary.

Q14. A relationship where one instructor can advise many students, but each student has at most one advisor is:

1:1

1:N from instructor to student

N:M

N:1 from instructor to student

Correct answer: B. The instructor side is one and the student side is many.

Q15. Every student must have an advisor, while an instructor may advise none. Student participation is:

Partial

Total

Weak

Derived

Correct answer: B. Every student participates at least once, so participation is total.

Q16. A candidate key is a:

Non-unique attribute

Minimal super key

Set of all attributes

Foreign key only

Correct answer: B. A candidate key uniquely identifies an entity and is minimal.

Q17. A super key may contain:

Only one attribute

No unique attributes

Extra attributes beyond those needed for uniqueness

Only foreign keys

Correct answer: C. A super key is unique but need not be minimal.

Q18. A strong entity set is one that:

Has no attributes

Has a primary key

Must be multivalued

Cannot participate in relationships

Correct answer: B. The slides define an entity set with a primary key as strong.

Q19. A weak entity is identified using:

Its partial key alone globally

The owner key plus its partial key

A derived attribute only

A relationship name only

Correct answer: B. Its key combines the identifying strong entity's key with its discriminator.

Q20. A many-to-many relationship is usually mapped to:

Only one entity table

A separate relation containing participating keys

A derived attribute

No relation

Correct answer: B. The relation stores the primary keys of both entity sets and any relationship attributes.

Easy Level — Fill in the Blanks

QE-F1. An object distinguishable from other objects is an _____.

Answer: entity
Explanation: This is the terminology used in the ER-model slides.

QE-F2. A collection of same-type entities is an entity _____.

Answer: set
Explanation: This is the terminology used in the ER-model slides.

QE-F3. The set of permitted values for an attribute is its _____.

Answer: domain
Explanation: This is the terminology used in the ER-model slides.

QE-F4. A _____ attribute has atomic, indivisible values.

Answer: simple
Explanation: This is the terminology used in the ER-model slides.

QE-F5. An attribute with component parts is _____.

Answer: composite
Explanation: This is the terminology used in the ER-model slides.

QE-F6. An attribute with one value for an entity is _____-valued.

Answer: single
Explanation: This is the terminology used in the ER-model slides.

QE-F7. Phone numbers are commonly modeled as a _____ attribute.

Answer: multivalued
Explanation: This is the terminology used in the ER-model slides.

QE-F8. Age computed from date of birth is a _____ attribute.

Answer: derived
Explanation: This is the terminology used in the ER-model slides.

QE-F9. A rectangle denotes an _____ set.

Answer: entity
Explanation: This is the terminology used in the ER-model slides.

QE-F10. A diamond denotes a _____ set.

Answer: relationship
Explanation: This is the terminology used in the ER-model slides.

QE-F11. An underlined attribute denotes a _____ key.

Answer: primary
Explanation: This is the terminology used in the ER-model slides.

QE-F12. A relationship involving two entity sets is _____.

Answer: binary
Explanation: This is the terminology used in the ER-model slides.

QE-F13. A relationship involving three entity sets is _____.

Answer: ternary
Explanation: This is the terminology used in the ER-model slides.

QE-F14. A minimal super key is a _____ key.

Answer: candidate
Explanation: This is the terminology used in the ER-model slides.

QE-F15. Every entity participating at least once means _____ participation.

Answer: total
Explanation: This is the terminology used in the ER-model slides.

QE-F16. Optional participation is called _____ participation.

Answer: partial
Explanation: This is the terminology used in the ER-model slides.

QE-F17. A weak entity uses a _____ diamond for its identifying relationship.

Answer: double
Explanation: This is the terminology used in the ER-model slides.

QE-F18. The weak entity discriminator is also called a _____ key.

Answer: partial
Explanation: This is the terminology used in the ER-model slides.

QE-F19. A 0..* constraint means zero or _____.

Answer: more
Explanation: This is the terminology used in the ER-model slides.

QE-F20. A separate relation is used to map an entity's _____ attribute.

Answer: multivalued
Explanation: This is the terminology used in the ER-model slides.

Easy Level — Short Application Questions

QE-A1. A college stores Student(RegNo, Name, Phone). Which attribute is the most natural primary key if RegNo is unique?

Answer: RegNo.
Explanation: It uniquely identifies each student and is more stable than a name or phone number.

QE-A2. A person has first name, middle initial and last name. Is Name simple or composite?

Answer: Composite.
Explanation: It can be decomposed into meaningful component attributes.

QE-A3. An instructor may advise zero or many students; every student must have exactly one advisor. State the cardinality and participation.

Answer: 1:N from Instructor to Student; Instructor partial, Student total.
Explanation: Instructor minimum is 0 and maximum many; Student minimum and maximum are 1.

QE-A4. A Course is identified by CourseID and has Title and Credits. Write the relation schema.

Answer: Course(CourseID, Title, Credits).
Explanation: A strong entity becomes a relation with the same attributes and its key.

QE-A5. A student takes many courses and each course has many students. Write the relationship relation.

Answer: Takes(StudentID, CourseID).
Explanation: For M:N, create a relation containing both entity keys; together they identify a pair if a student takes a course only once.

Medium Level — MCQs

Q1. Instructor(ID, Name, Email) has unique ID and unique Email. Which is a candidate key set?

{ID, Name} only

{ID}, {Email}

{Name} only

{ID, Name, Email} only

Correct answer: B. ID and Email are each minimal unique identifiers; the larger combinations are not minimal.

Q2. If (ID, Name) uniquely identifies instructors but ID alone is unique, then (ID, Name) is:

A candidate key

A super key but not a candidate key

Not a super key

A partial key

Correct answer: B. It is unique but not minimal because ID alone is sufficient.

Q3. An employee may work in at most one department; a department may have many employees. The relationship is:

1:1

1:N from department to employee

N:M

Ternary

Correct answer: B. One department relates to many employees; each employee maps to at most one department.

Q4. A student can take many subjects and a subject can be taken by many students. The relationship is:

1:1

1:N

N:1

M:N

Correct answer: D. Both directions allow multiple associated entities.

Q5. In a 1:N relationship, where should the foreign key normally go?

On the 1 side

On the N side

In a separate relation always

In neither table

Correct answer: B. The N-side relation stores the key of the 1-side relation.

Q6. For a 1:1 relationship, the foreign key may be placed:

Only on the alphabetically first entity

On either participating relation, subject to constraints

Only in a third relation

Only on the weak entity

Correct answer: B. Either side can be selected to carry the other side's key.

Q7. For an M:N relationship Student–Course, which relation is appropriate?

Student(StudentID, CourseID) only

Course(StudentID, CourseID) only

Enrollment(StudentID, CourseID)

No relation needed

Correct answer: C. A separate relation represents the M:N association.

Q8. A relationship has attribute Advisor_On_Date. The date describes:

Only the instructor

Only the student

The advisor association

The domain of ID

Correct answer: C. The date has meaning for the instructor–student association.

Q9. A line marked 0..* means:

Exactly one

At least one and at most one

Zero or more

One or more only

Correct answer: C. Minimum is zero and maximum is unbounded.

Q10. A line marked 1..1 means:

Optional participation

Exactly one

Many or none

At least two

Correct answer: B. Minimum and maximum are both one.

Q11. If some instructors do not advise any student, instructor participation in advisor is:

Total

Partial

Identifying

Ternary

Correct answer: B. Some instructors may participate zero times.

Q12. A weak entity's partial key distinguishes:

All weak entities in the entire database

Weak entities belonging to a particular owner

Owners from each other

Relationships only

Correct answer: B. The discriminator is unique within the owner's set of weak entities.

Q13. Why is a separate relation for Section–Course identifying relationship often omitted?

Course has no key

Section already includes Course's key

Section has no attributes

It is always M:N

Correct answer: B. The owner key is already part of the weak entity's relation.

Q14. A multivalued phone_number of Instructor is mapped as:

Instructor(ID, phone1, phone2)

InstructorPhone(ID, phone_number)

Instructor(phone_number) only

Phone(ID) only

Correct answer: B. A separate relation stores owner key and one phone value per row.

Q15. Composite Name(first, middle, last) is mapped by:

Keeping a nested Name object only

Flattening its components into separate columns

Making Name a foreign key

Discarding the components

Correct answer: B. Composite attributes are flattened into component attributes.

Q16. In a recursive Works_For(Employee) relationship, manager_id is a foreign key to:

Department ID

Employee.employee_id

Salary

A new unrelated table

Correct answer: B. The relationship connects employees to other employees in different roles.

Q17. A relationship among Employee, Project, and Supplier has degree:

Unary

Binary

Ternary

Not a relationship

Correct answer: C. Three participating entity sets make it ternary.

Q18. If every loan must be associated with a customer, Loan participation is:

Partial

Total

Optional

Derived

Correct answer: B. No loan entity can exist without participating in Borrower.

Q19. If some customers have never taken a loan, Customer participation in Borrower is:

Total

Partial

Weak

Identifying

Correct answer: B. Customers without loans do not participate.

Q20. The key of an M:N relationship without a descriptive attribute is generally:

Only the first entity key

Only the second entity key

Combination of participating entity primary keys

A random generated value required by ER theory

Correct answer: C. The slides specify the union/combination of the participating primary keys.

Medium Level — Fill in the Blanks

QM-F1. In a 1:N relationship, the foreign key is usually placed on the _____ side.

Answer: many / N
Explanation: Use the rule with the stated relationship constraint.

QM-F2. An M:N relationship is represented by a separate relation containing keys from _____ participating entities.

Answer: both
Explanation: Use the rule with the stated relationship constraint.

QM-F3. A composite attribute is mapped by flattening it into its _____ attributes.

Answer: component
Explanation: Use the rule with the stated relationship constraint.

QM-F4. The owner's key plus the partial key forms the _____ key of a weak entity.

Answer: primary
Explanation: Use the rule with the stated relationship constraint.

QM-F5. A recursive relationship has the same entity set participating in different _____.

Answer: roles
Explanation: Use the rule with the stated relationship constraint.

QM-F6. A degree-n relationship is called _____-ary.

Answer: n
Explanation: Use the rule with the stated relationship constraint.

QM-F7. A relationship attribute describes the _____ rather than either entity alone.

Answer: association
Explanation: Use the rule with the stated relationship constraint.

QM-F8. In the notation l..h, l denotes the _____ cardinality.

Answer: minimum
Explanation: Use the rule with the stated relationship constraint.

QM-F9. In the notation l..h, h denotes the _____ cardinality.

Answer: maximum
Explanation: Use the rule with the stated relationship constraint.

QM-F10. A double line indicates _____ participation.

Answer: total
Explanation: Use the rule with the stated relationship constraint.

QM-F11. A single line commonly indicates _____ participation in Chen notation.

Answer: partial
Explanation: Use the rule with the stated relationship constraint.

QM-F12. A multivalued attribute M of entity E is mapped to a separate relation containing E's key and _____.

Answer: M / the multivalued value
Explanation: Use the rule with the stated relationship constraint.

QM-F13. The relationship between Employee and Department is many-to-one from Employee to _____.

Answer: Department
Explanation: Use the rule with the stated relationship constraint.

QM-F14. A foreign key references a key in another relation or in the _____ relation for a recursive relationship.

Answer: same
Explanation: Use the rule with the stated relationship constraint.

QM-F15. A relationship set's key can include the keys of all participating _____ sets.

Answer: entity
Explanation: Use the rule with the stated relationship constraint.

QM-F16. A section's SecID may only be unique within a course; SecID is then a _____ key.

Answer: partial
Explanation: Use the rule with the stated relationship constraint.

QM-F17. A weak entity's identifying relationship is represented by a _____ diamond.

Answer: double
Explanation: Use the rule with the stated relationship constraint.

QM-F18. For 1:1, either side may carry the foreign key, provided the mapping enforces _____.

Answer: uniqueness
Explanation: Use the rule with the stated relationship constraint.

QM-F19. If a customer may exist without taking a loan, Customer participation is _____.

Answer: partial
Explanation: Use the rule with the stated relationship constraint.

QM-F20. If every loan must belong to at least one customer, Loan participation is _____.

Answer: total
Explanation: Use the rule with the stated relationship constraint.

Medium Level — Short Application Questions

QM-A1. Schema: Employee(EmpID PK, Name, DeptID FK). Department(DeptID PK, DeptName). What relationship does this suggest?

Answer: Department 1:N Employee (or Employee N:1 Department).
Explanation: Many Employee rows can reference one Department; each employee row points to at most one department.

QM-A2. Schema: Student(StudentID PK), Course(CourseID PK), Enrollment(StudentID FK, CourseID FK, Grade). Identify the relationship and key.

Answer: Student–Course is M:N; Enrollment's usual composite key is (StudentID, CourseID), with Grade as a relationship attribute.
Explanation: Grade belongs to a specific student-course pair. If repeat attempts are allowed, add an attempt/term discriminator.

QM-A3. A weak entity Payment has PaymentNo unique only within a Loan. What is its key?

Answer: (LoanNo, PaymentNo).
Explanation: PaymentNo is a partial key; LoanNo identifies the owning loan.

QM-A4. A customer's address contains street, city, state and zip. How is it mapped?

Answer: Flatten into separate columns such as Street, City, State, Zip.
Explanation: The slide rule maps composite attributes to their components.

QM-A5. A professor supervises other professors. What type of relationship is this?

Answer: Recursive relationship.
Explanation: The Professor entity set participates in the same relationship in supervisor and supervisee roles.

Hard Level — MCQs

Q1. A Department has many Employees; each Employee works in at most one Department. Some employees may not yet be assigned. Which is correct?

Department total; Employee total necessarily

Department 1:N Employee; Employee participation may be partial

M:N with a mandatory junction table

Employee weak entity

Correct answer: B. Cardinality is 1:N from department to employee; optional assignment means employee participation is partial.

Q2. Each project must belong to exactly one department, and a department may offer zero or more projects. Which min–max constraints fit?

Department 1..1, Project 0..*

Department 0..*, Project 1..1

Both 0..*

Both 1..1

Correct answer: B. A department may have no projects; each project has exactly one department.

Q3. A section is identified by CourseID plus (SectionID, Semester, Year). SectionID may repeat across different courses. What is the key?

SectionID

CourseID only

(CourseID, SectionID, Semester, Year)

(Semester, Year)

Correct answer: C. The owner key plus the discriminator and remaining distinguishing attributes form the full key.

Q4. For a Student–Course M:N relationship with attribute Grade, the best mapping is:

Student(StudentID, Grade)

Course(CourseID, Grade)

Takes(StudentID, CourseID, Grade)

Grade(Grade)

Correct answer: C. Grade describes a particular student taking a particular course.

Q5. A student may take a course multiple times in different semesters. Why may (StudentID, CourseID) be insufficient as relationship key?

StudentID is not a key

CourseID is not a key

The same pair can recur; include a distinguishing attempt/semester attribute

M:N cannot have keys

Correct answer: C. If repeated enrollments are allowed, another attribute must distinguish each occurrence; this is an assumption beyond the simple one-pair example.

Q6. A multivalued attribute Emails belongs to Customer. Which mapping preserves multiple emails without repeating customer columns?

Customer(CustomerID, Email1, Email2)

CustomerEmail(CustomerID, Email)

Customer(Email)

Email is derived

Correct answer: B. One row per email value avoids fixed repeated columns.

Q7. A relationship is shown as total participation on Employee, but the written requirement says employees may be unassigned. What should the modeler do?

Ignore the wording

Flag the contradiction and state an assumption

Convert Employee to weak

Make the relationship ternary

Correct answer: B. The diagram and requirement conflict; resolve explicitly rather than silently choosing.

Q8. Which claim about total participation is true?

Only weak entities can have total participation

A strong entity may have total participation

Total participation means 1:1

Total participation means maximum is many

Correct answer: B. Total participation is independent of strong/weak status.

Q9. A weak entity Dependent has partial key DependentName and owner Employee(EmpID). What is the relational primary key?

DependentName

EmpID only

(EmpID, DependentName)

(EmpID, EmployeeName)

Correct answer: C. The owner's primary key is combined with the weak entity discriminator.

Q10. For a 1:1 relationship where every Institute has exactly one Head and every Head belongs to exactly one Institute, which mapping is valid?

Put InstituteID as unique FK in Head, or HeadID as unique FK in Institute

Always use an M:N table

Place both foreign keys in both tables without constraints

Drop the relationship

Correct answer: A. Either side can carry the other's key, with uniqueness and participation constraints reflected.

Q11. A 1:N relationship has descriptive attribute StartDate. Where should StartDate be mapped?

Always on the 1-side

On the N-side with its FK when one relationship per N-side entity is possible

In an unrelated table in every case

As a multivalued attribute automatically

Correct answer: B. For a 1:N relationship, relationship attributes can usually be stored on the N-side alongside the FK.

Q12. A ternary relationship Supply(Supplier, Part, Project) records Quantity per supplier-part-project combination. Why is three binary relationships not automatically equivalent?

Binary relationships cannot have keys

The ternary association captures a combination-specific fact

Ternary relationships have no attributes

All relationships must be 1:1

Correct answer: B. Quantity depends on the full triple; binary splits may lose which combination the fact belongs to.

Q13. Given Enrollment(StudentID FK, CourseID FK, Grade), which is the likely relationship cardinality between Student and Course?

1:1

1:N only

M:N

Recursive

Correct answer: C. Many students may enroll in many courses; Enrollment resolves the M:N relationship.

Q14. An entity has ID and Email, both individually unique. If the organization uses ID as the official identifier, then:

Email cannot be a candidate key

ID is primary key and Email can remain an alternate candidate key

ID and Email together are the only key

Email is a foreign key automatically

Correct answer: B. One candidate key is chosen as primary; the other remains an alternate candidate key.

Q15. A schema contains Section(CourseID, SecID, Semester, Year) and a separate Sec_Course with the same columns representing the identifying relationship. What is the concern?

Missing attribute domain

Redundancy

No primary key possible

Wrong degree

Correct answer: B. The owner key already appears in Section, so a duplicate relation may be redundant.

Q16. An employee has a manager who is also an employee. Which schema can encode this recursive relationship?

Employee(EmpID, Name, ManagerID FK→Employee.EmpID)

Manager(ManagerID) only

Employee(EmpID, Name), Department(ManagerID)

Employee(EmpID, Name, ManagerName without key)

Correct answer: A. ManagerID references another row in the same Employee relation.

Q17. A project can receive funds from many agencies; an agency can fund many projects. GrantOrderNo and GrantYear describe each funding award. What should be modeled?

Project–Agency 1:1

Project–Agency M:N with award attributes on the association or award entity

Agency as a derived attribute

Project as a weak entity necessarily

Correct answer: B. The scenario is M:N and grant details belong to a particular funding association/award.

Q18. Which is the most defensible conclusion from a foreign key alone?

The relationship is always total on both sides

The referencing row points to a referenced row, but optionality needs more constraints

The relationship is always M:N

The foreign key is always a primary key

Correct answer: B. A foreign key expresses referential linkage; nullability/participation and uniqueness determine more.

Q19. A relationship between Doctor, Patient, and Hospital records the consultation location and participants together. If the event is specifically one consultation involving all three, what degree is natural?

Unary

Binary only

Ternary

Weak

Correct answer: C. The relationship instance is defined by three participating entity sets.

Q20. A relationship key is described as the union of participating entity keys, but the relationship is 1:N. What is the nuance?

The union is always the minimal key

Cardinality may allow a smaller key for the relationship; the simple union is a general identifying combination

Relationship sets cannot have keys

The entity keys are ignored

Correct answer: B. The slide gives the combined keys as the relationship identifier; under functional/cardinality constraints, a subset can sometimes suffice.

Hard Level — Fill in the Blanks

QH-F1. If a scenario says an entity may participate zero times, its minimum participation is _____.

Answer: 0
Explanation: The answer follows from the scenario and the ER mapping rules.

QH-F2. If every entity must participate at least once, the minimum is _____.

Answer: 1
Explanation: The answer follows from the scenario and the ER mapping rules.

QH-F3. A weak entity is existence-dependent on its _____ entity set.

Answer: identifying / strong owner
Explanation: The answer follows from the scenario and the ER mapping rules.

QH-F4. A ternary relationship relates _____ entity sets in one relationship instance.

Answer: three
Explanation: The answer follows from the scenario and the ER mapping rules.

QH-F5. A relationship attribute such as Quantity in Supply depends on the participating entity _____.

Answer: combination
Explanation: The answer follows from the scenario and the ER mapping rules.

QH-F6. When a 1:N relationship is mapped by an FK on the N-side, the FK references the key on the _____ side.

Answer: 1 / one
Explanation: The answer follows from the scenario and the ER mapping rules.

QH-F7. A relationship schema may be omitted for a weak entity's identifying relationship when the owner key is already included in the weak entity's _____.

Answer: relation/schema
Explanation: The answer follows from the scenario and the ER mapping rules.

QH-F8. A unique but non-minimal set of attributes is a _____ key but not a candidate key.

Answer: super
Explanation: The answer follows from the scenario and the ER mapping rules.

QH-F9. If both sides of a 1:1 relationship must correspond exactly once, both sides have _____ participation.

Answer: total
Explanation: The answer follows from the scenario and the ER mapping rules.

QH-F10. The Employee.ManagerID self-reference points to Employee's _____ key.

Answer: primary
Explanation: The answer follows from the scenario and the ER mapping rules.

QH-F11. The key of a weak entity includes the owner key and its _____ key.

Answer: partial
Explanation: The answer follows from the scenario and the ER mapping rules.

QH-F12. In a relationship with an attribute Date, the date should be stored where it describes the _____.

Answer: relationship/association
Explanation: The answer follows from the scenario and the ER mapping rules.

QH-F13. A foreign key does not by itself guarantee total participation; the minimum participation constraint must also be _____.

Answer: specified/enforced
Explanation: The answer follows from the scenario and the ER mapping rules.

QH-F14. If a model diagram conflicts with the written requirement, the modeler should record a clear _____.

Answer: assumption
Explanation: The answer follows from the scenario and the ER mapping rules.

QH-F15. A relationship between three entity sets has degree _____.

Answer: 3
Explanation: The answer follows from the scenario and the ER mapping rules.

QH-F16. The union of the primary keys of participating entities is a general identifier for a _____ instance.

Answer: relationship
Explanation: The answer follows from the scenario and the ER mapping rules.

QH-F17. For multivalued Phone, each phone value is represented by a separate _____.

Answer: tuple/row
Explanation: The answer follows from the scenario and the ER mapping rules.

QH-F18. For a 1:1 mapping, the chosen foreign key should be constrained as _____ to prevent multiple matches.

Answer: unique
Explanation: The answer follows from the scenario and the ER mapping rules.

QH-F19. An employee may have no manager at the top of an organization; ManagerID can be _____.

Answer: null / absent
Explanation: The answer follows from the scenario and the ER mapping rules.

QH-F20. A relationship that permits multiple students per course and multiple courses per student is _____-to-_____.

Answer: many-to-many
Explanation: The answer follows from the scenario and the ER mapping rules.

Hard Level — Multi-step Modeling Questions

QH-A1. A university stores Course(CourseID, Title) and Section(SecID, Semester, Year). SecID repeats across courses. Give the Section key and relationship.

Answer: Section(CourseID, SecID, Semester, Year); Course 1:N Section through an identifying relationship.
Explanation: Section is weak relative to Course if its own attributes cannot uniquely identify it globally. The Course key plus section discriminator/term identifies it.

QH-A2. An agency may fund many projects; a project may receive funds from many agencies. Each award has GrantOrderNo, Year and Amount. Propose a model.

Answer: Agency and Project have an M:N Funds relationship or an Award associative entity with GrantOrderNo, Year, Amount.
Explanation: Award details describe one particular agency-project funding event, not Agency or Project alone. If a grant order uniquely identifies an award, it may be the award key.

QH-A3. A clinic visit connects one Doctor, one Patient and one Clinic and records VisitDate. Which relationship degree and mapping are appropriate?

Answer: Ternary Visit(DoctorID FK, PatientID FK, ClinicID FK, VisitDate); choose a key based on whether repeat visits can occur.
Explanation: The fact concerns the combination of doctor, patient and clinic. If repeat visits are possible, VisitDate/time or a VisitID is needed to distinguish events.

QH-A4. An Employee may manage many Employees; each employee has at most one manager. Model it.

Answer: Employee(EmpID PK, Name, ManagerID FK → Employee.EmpID).
Explanation: This is a recursive 1:N relationship; ManagerID is optional for top-level employees.

QH-A5. A Student may have several phone numbers. Show a normalized mapping.

Answer: Student(StudentID PK, Name); StudentPhone(StudentID FK, PhoneNumber; PK(StudentID, PhoneNumber)).
Explanation: Each phone is a separate tuple; do not add Phone1/Phone2 columns.

QH-A6. A company requires every employee to belong to one department, but a department may have zero employees. State participation and schema.

Answer: Department 1:N Employee; Employee total, Department partial. Employee(EmpID PK, Name, DeptID FK NOT NULL).
Explanation: Every employee must participate; a department can exist with no employees. The FK is on the many side.

ER-to-Relational-Schema Conversion Practice

Use the source slides' mapping rules: strong entity → relation; 1:N → FK on N-side; 1:1 → FK on either side with uniqueness; M:N → separate relation with both keys and relationship attributes; composite attributes → component columns; multivalued attributes → separate relation; weak entity → owner key plus partial key.

Conversion 1 — Student Takes Course (M:N)

ER description: Student(StudentID, Name) and Course(CourseID, Title, Credits) are connected by Takes; Takes has attribute Grade. A student may take many courses and a course may be taken by many students.

```
STUDENT( StudentID [PK], Name )
COURSE( CourseID [PK], Title, Credits )
TAKES( StudentID [PK, FK -> STUDENT.StudentID],
       CourseID [PK, FK -> COURSE.CourseID], Grade )
```

Reasoning: M:N cannot be represented by one foreign key on only one side. Takes stores both primary keys. The pair is the key if each student-course pair occurs once; if repeated attempts are allowed, include AttemptNo or Term in the key.

Conversion 2 — Department Employs Employee (1:N)

ER description: A department has zero or more employees; each employee works in exactly one department.

```
DEPARTMENT( DeptNo [PK], DeptName )
EMPLOYEE( EmpNo [PK], EmpName, Salary, DeptNo [FK -> DEPARTMENT.DeptNo] )
```

Reasoning: Department is the 1-side and Employee is the N-side. Put DeptNo in Employee. Because every employee must belong to a department, DeptNo is mandatory/non-null. Department participation is partial because a department may have no employees.

Conversion 3 — Institute Has Head (1:1)

```
INSTITUTE( InstName [PK], Location, HeadID [FK, UNIQUE -> HEAD.HeadID] )
HEAD( HeadID [PK], Name, Qualification )
```

Reasoning: Either side can carry the foreign key for a 1:1 relationship. Here HeadID is placed in Institute and constrained unique. If every institute must have a head, the FK must also be non-null. The exact side chosen is a design choice.

Conversion 4 — Composite and Derived Attributes

ER description: Instructor has ID, composite Name(First, Middle, Last), Address(StreetNo, StreetName, City, State, Zip), DateOfBirth, and derived Age.

```
INSTRUCTOR( ID [PK], FirstName, MiddleInitial, LastName,
            StreetNo, StreetName, City, State, Zip, DateOfBirth )
```

Reasoning: Flatten the composite attributes into component columns. Age is derived from DateOfBirth and is normally not stored if it can be computed reliably.

Conversion 5 — Multivalued Phone

```
INSTRUCTOR( ID [PK], Name )
INSTRUCTOR_PHONE( ID [PK, FK -> INSTRUCTOR.ID],
                  PhoneNumber [PK] )
```

Reasoning: Each phone number becomes a separate row. The composite primary key (ID, PhoneNumber) prevents duplicate phone values for the same instructor.

Conversion 6 — Weak Entity Section

ER description: Course(CourseID, Title, Credits) owns Sections. SecID, Semester and Year distinguish sections for a given course.

```
COURSE( CourseID [PK], Title, Credits )
SECTION( CourseID [PK, FK -> COURSE.CourseID],
         SecID [PK], Semester [PK], Year [PK] )
```

Reasoning: Section's primary key includes the owner key and its discriminator/identifying attributes. The separate identifying relation can be omitted because CourseID is already included in SECTION.

Conversion 7 — Recursive Employee Works-For

```
EMPLOYEE( EmployeeID [PK], Name,
          ManagerID [FK -> EMPLOYEE.EmployeeID, nullable] )
```

Reasoning: The same entity set appears in worker and manager roles. ManagerID references another employee row; it may be null for a top-level manager if the requirements allow this.

Conversion 8 — Banking Example from the Slides

```
ACCOUNT( Ac_No [PK], Balance, BranchName [FK -> BRANCH.BranchName] )
BRANCH( BranchName [PK], BranchCity, Assets )
CUSTOMER( CustomerName [PK], CustomerCity, CustomerStreet )
LOAN( LoanNo [PK], Amount, BranchName [FK -> BRANCH.BranchName] )
DEPOSITOR( Ac_No [PK, FK -> ACCOUNT.Ac_No],
           CustomerName [PK, FK -> CUSTOMER.CustomerName] )
BORROWER( CustomerName [PK, FK -> CUSTOMER.CustomerName],
          LoanNo [PK, FK -> LOAN.LoanNo] )
```

Reasoning: Depositor and Borrower are M:N association relations in the slide's listed schema. Branch-to-Account and Branch-to-Loan are modeled with the branch key on the many side. Assumption: BranchName and CustomerName are treated as keys because that is how the slide's abbreviated schema is written; in a production design, stable IDs are usually safer.

Conversion 9 — Employee Works on Project

```
DEPARTMENT( DeptNo [PK], DeptName, Location )
PROJECT( ProjectNo [PK], ProjectName, Budget )
EMPLOYEE( EmployeeNo [PK], LastName, FirstName, DeptNo [FK -> DEPARTMENT.DeptNo] )
WORKS_ON( ProjectNo [PK, FK -> PROJECT.ProjectNo],
          EmployeeNo [PK, FK -> EMPLOYEE.EmployeeNo],
          EnterDate, Job )
```

Reasoning: Employee-to-Department is N:1, so DeptNo is placed in Employee. Employee–Project is M:N; Works_On stores both keys and the relationship attributes EnterDate and Job.

Conversion 10 — Professor and Sponsored Project

```
PROFESSOR( EmpNo [PK], Name, Address )
SPONSORED_PROJECT( ProjectID [PK], Name, Duration )
WORKS_FOR( EmpNo [PK, FK -> PROFESSOR.EmpNo],
           ProjectID [PK, FK -> SPONSORED_PROJECT.ProjectID],
           PercentTime )
```

Reasoning: The relationship is M:N when professors can work on multiple projects and projects can involve multiple professors. PercentTime belongs to each professor-project association.

Relational-Schema-to-ER-Model Conversion Practice

A relational schema does not always reveal every business rule. Foreign keys suggest relationships; unique constraints help identify 1:1 relationships; repeated foreign-key values normally support 1:N; a junction table with two foreign keys usually indicates M:N. Participation constraints require NOT NULL/other constraints or a written business rule.

Schema-to-ER 1 — Employee and Department

```
DEPARTMENT(DeptNo PK, DeptName)
EMPLOYEE(EmpNo PK, Name, Salary, DeptNo FK NOT NULL)
```

ER solution: Entities Department(DeptNo, DeptName) and Employee(EmpNo, Name, Salary). Relationship Works_In is 1:N from Department to Employee. Employee participation is total because DeptNo is NOT NULL; Department participation is partial unless a rule requires every department to have an employee.

Schema-to-ER 2 — Student and Course Enrollment

```
STUDENT(StudentID PK, Name)
COURSE(CourseID PK, Title)
ENROLLMENT(StudentID PK/FK, CourseID PK/FK, Grade)
```

ER solution: Student and Course are entities. Enrollment represents M:N relationship Takes with descriptive attribute Grade. Both sides are usually partial unless requirements require every student/course to be involved in at least one enrollment.

Schema-to-ER 3 — Recursive Employee

```
EMPLOYEE(EmployeeID PK, Name, ManagerID FK -> EMPLOYEE.EmployeeID)
```

ER solution: One Employee entity set participates twice in Manages/Reports_To in the roles Manager and Worker. A manager may have many workers; each worker has at most one manager. Manager participation is partial; worker-side participation depends on whether top-level employees are allowed.

Schema-to-ER 4 — Project Funding

```
AGENCY(GrantOrderNo PK, AgencyName, ContactPerson, Email, Phone, TotalGrant, GrantYear)
PROJECT(ProjectID PK, Title, Duration)
FUNDING(GrantOrderNo PK/FK, ProjectID PK/FK, Amount)
```

ER solution: Agency and Project are entities connected by M:N Funds. Funding stores the association-specific amount. If a grant order can fund multiple projects or has a separate award identity, revise the key based on the actual business rule; do not infer more than the schema supports.

Schema-to-ER 5 — Customer Phones

```
CUSTOMER(CustomerID PK, Name)
CUSTOMER_PHONE(CustomerID PK/FK, PhoneNumber PK)
```

ER solution: Customer has a multivalued PhoneNumber attribute. The separate relation is the standard relational representation of that multivalued attribute. The relation alone does not tell whether at least one phone is mandatory.

Schema-to-ER 6 — Weak Dependent

```
EMPLOYEE(EmpNo PK, Designation)
DEPENDENT(EmpNo PK/FK, DependentName PK, Relation, Age)
```

ER solution: Employee is strong; Dependent is weak. Depends_On is identifying and 1:N from Employee to Dependent. Dependent participation is total; Employee participation may be partial. DependentName is a partial key, and (EmpNo, DependentName) is the relational primary key.

Schema-to-ER 7 — Banking

```
BRANCH(BranchName PK, City, Assets)
ACCOUNT(AccountNo PK, Balance, BranchName FK)
CUSTOMER(CustomerID PK, Name)
DEPOSITOR(CustomerID PK/FK, AccountNo PK/FK)
```

ER solution: Branch–Account is 1:N if each account references at most one branch. Customer–Account is M:N through Depositor. If the schema uses NOT NULL on BranchName, account participation is total; otherwise the FK alone may allow optional assignment.

How to infer constraints cautiously

• A foreign key without UNIQUE commonly supports N:1 from referencing relation to referenced relation.

• A foreign key with UNIQUE can support 1:1, but nullability still affects participation.

• A junction relation whose primary key combines two foreign keys usually represents M:N.

• A self-referencing foreign key suggests a recursive relationship.

• A table whose primary key contains an owner FK plus a local discriminator may represent a weak entity.

• The schema alone may not reveal whether participation is total if nullability and business rules are omitted.

Original Tutorial Questions and Solutions from the Reference

The final slide of the supplied PDF lists five tutorial questions explicitly. The answers below preserve their meaning and show the reasoning. The PDF also contains labeled ER exercises and worked conversion slides; these are handled in the preceding conversion sections where the schemas are legible.

Tutorial 1 — Explain the difference between a weak and a strong entity set.

QT1. Explain the difference between a weak and a strong entity set.

> **Answer:** A strong entity set has its own primary key that identifies each entity. A weak entity set does not have sufficient attributes to form a complete primary key on its own and depends on an identifying strong entity.
> **Explanation:** A weak entity is identified by combining the owner's primary key with the weak entity's partial key. In the slides, Section depends on Course, and Dependent depends on Employee.

Tutorial 2 — Distinguish primary key, candidate key, and super key with an example.

QT2. Explain the distinctions among primary key, candidate key, and super key with an example.

> **Answer:** For Instructor(ID, Name, Email), assume ID and Email are each unique. Super key: any attribute set that uniquely identifies a row, such as {ID}, {Email}, or {ID, Name}. Candidate keys: minimal super keys {ID} and {Email}. Primary key: the candidate key chosen as the official identifier, for example {ID}.
> **Explanation:** {ID, Name} is a super key but not a candidate key because {ID} alone is already unique. The slides explicitly use ID and Email as candidate keys and select ID semantically as primary key.

Tutorial 3 — A weak entity can be made strong by adding the owner key. What redundancy results?

QT3. A weak entity set can always be made into a strong entity set by adding the primary-key attributes of its identifying entity set. Outline the redundancy that results if this is done.

> **Answer:** If the owner's key is stored as an ordinary identifying attribute in a supposedly strong child while the identifying relationship is also modeled separately, the same owner-child association is represented twice.
> **Explanation:** For example, if Section stores CourseID and a separate Sec_Course relation also stores CourseID and Section's key, the identifying link is duplicated. The slides note that the separate sec_course schema is redundant because Section already contains CourseID.

Tutorial 4 — Is total participation possible for a strong entity? Give an example.

QT4. In order to have total participation, does an entity have to be weak? If not, give a strong entity participating in total participation.

> **Answer:** No. Total participation is independent of whether an entity is strong or weak. Example: Student is a strong entity with StudentID as its primary key, but every Student may be required to have an Advisor; therefore Student participates totally in Advisor.
> **Explanation:** A strong entity can have a key of its own and still be subject to a mandatory relationship. The slide example explicitly shows Student total participation in advisor.

Tutorial 5 — Car-insurance company ER model

QT5. Construct an ER diagram for a car-insurance company whose customers own one or more cars each. Each car has associated with it zero to any number of recorded accidents.

Answer: Entities: CUSTOMER(CustomerID [PK], Name, Address, Phone); CAR(VIN [PK], Make, Model, Year); ACCIDENT(AccidentID [PK], AccidentDate, Location, Description). Relationships: OWNS between Customer and Car is 1:N from Customer to Car, with total participation on Customer if every customer in the modeled set must own at least one car; ACCIDENT_INVOLVEMENT between Car and Accident is M:N if an accident may involve multiple cars and each car may have zero or more accidents.
Explanation: The prompt says each customer owns one or more cars, so Customer minimum participation is 1 and maximum is many. It says each car has zero to any number of recorded accidents, so Car participation in the accident relationship is partial (0..*). Assumption: a recorded accident can involve multiple cars; if the course expects each accident to be linked to exactly one car, the Accident–Car cardinality should instead be 1:N. A relational mapping for the stated multi-car assumption is OWNS(CustomerID, VIN) if a car can be co-owned or CAR(CustomerID FK) if each car has one owner; ACCIDENT_INVOLVEMENT(VIN, AccidentID, DamageDetails optional).

```
CUSTOMER( CustomerID [PK], Name, Address, Phone )
CAR( VIN [PK], Make, Model, Year )
ACCIDENT( AccidentID [PK], AccidentDate, Location, Description )
OWNS( CustomerID [PK, FK -> CUSTOMER.CustomerID], VIN [PK, FK -> CAR.VIN] )
ACCIDENT_INVOLVEMENT( VIN [PK, FK -> CAR.VIN],
                      AccidentID [PK, FK -> ACCIDENT.AccidentID] )
```

Diagram representation: CUSTOMER —(1:N, Customer total)→ OWNS ←(N:1)— CAR; CAR —(M:N, Car minimum 0)↔ ACCIDENT through ACCIDENT_INVOLVEMENT. In a hand-drawn Chen diagram, use rectangles for entities, diamonds for relationships, underline primary keys, and double-line Customer participation if every customer must own a car.

Other Source Examples and Conversion Exercises

The following compact solutions cover additional clearly readable examples and schema-conversion slides in the reference.

Employee–Department example (slides around pp. 35–39)

Entities: EMP(EmpNo [PK], Ename, Sal) and DEPT(DeptNo [PK], Dname). Works_In is N:1 from Employee to Department; each employee works in at most one department. Schema: EMP(EmpNo [PK], Ename, Sal, DeptNo [FK]) and DEPT(DeptNo [PK], Dname). The slides caution against redundantly modeling the same association twice.

Department offers Course (slides around pp. 37–39)

Entities: DEPT(DeptNo [PK], Dname) and COURSE(CourseID [PK], CourseName, NoOfSem, TotalCredits). Offers is 1:N from Department to Course if each course belongs to one department and a department can offer many courses. Map DeptNo as an FK in COURSE. If courses can be offered by multiple departments, the requirement changes to M:N and needs OFFERS(DeptNo, CourseID).

Student–Subject example (slides around pp. 33–34)

STUDENT(RegNo [PK], Name, Phone), SUBJECT(SubjectID [PK], SubjectName, Credits). Takes is M:N. Mapping: TAKES(RegNo [PK, FK], SubjectID [PK, FK]).

Professor–Course example from the conversion slides

The slide gives Professor(Name, Phone, Address, Sex, CourseID) and Course(CourseID, Name, Credits), with Professor.CourseID referencing Course.CourseID. This is consistent with a many-professors-to-one-course relationship if each professor row stores at most one course. Preserve the slide's schema for exam practice; if the intended business rule is professors teach multiple courses and courses have multiple professors, use a separate M:N relation instead.

Professor–Department example from the conversion slides

Professor(Name, Phone, Address, Sex, Dep_Name) and Department(Name, Location), with Dep_Name as a foreign key to Department.Dep_Name. This models many professors to one department, assuming each professor belongs to at most one department.

Employee–Dependent example from the conversion slides

EMPLOYEE(EmpNo [PK], EName) and MEDICARE_DEPENDENTS(EmpNo [PK, FK], Depnd_Name [PK], Relationship_With). EmpNo in the dependent relation references Employee.EmpNo. This is a weak entity mapping with owner key plus partial key.

Schema diagram reminders

A schema diagram draws each relation as a box, places the relation name at the top, lists attributes inside, underlines primary keys, and draws arrows from foreign-key attributes to the referenced primary-key attributes.

Quick Revision Sheet

Core definitions

• **Entity:** distinguishable real-world object.

• **Entity set:** collection of same-type entities with the same properties.

• **Attribute domain:** permitted values for an attribute.

• **Relationship:** association among entities; relationship set is a collection of same-type associations.

• **Relationship degree:** number of participating entity sets — binary (2), ternary (3), n-ary (n).

• **Descriptive relationship attribute:** a property of the association itself, such as Advisor_On_Date or Grade.

• **Super key:** any attribute set that uniquely identifies an entity.

• **Candidate key:** minimal super key.

• **Primary key:** selected candidate key.

• **Strong entity:** has its own primary key.

• **Weak entity:** depends on an owner; key = owner key + partial key.

Notation

Symbol / convention	Meaning
Rectangle	Entity set
Diamond	Relationship set
Oval	Attribute (including relationship attributes when attached to a diamond)
Underlined attribute	Primary key
Dashed underline	Weak entity partial key / discriminator
Double rectangle	Weak entity set
Double diamond	Identifying relationship
Double line	Total participation
Single line	Partial participation in the Chen notation used in the slides
Arrow / directed line	“One” side in the slide’s cardinality notation
Undirected line	“Many” side in the slide’s cardinality notation

Cardinality and participation

Type	Meaning / example
1:1	Each side associated with at most one on the other side.
1:N	One entity on A can associate with many on B; each B maps to at most one A.
N:1	Many on A map to at most one on B.
M:N	Many entities on either side can be associated.
0..*	Zero or more; optional / partial minimum.
1..1	Exactly one; mandatory / total on that role.
Total participation	Every entity participates in at least one relationship instance.
Partial participation	Some entities may participate zero times.

Mapping rules to memorize

• Strong entity → relation with its attributes and primary key.

• Weak entity → relation containing owner primary key plus partial key; owner key is part of the weak entity's primary key.

• 1:N or N:1 → put the one-side primary key as an FK on the many side.

• 1:1 → put an FK on either side; enforce uniqueness and mandatory participation as required.

• M:N → create a separate relation with keys from both sides plus relationship attributes.

• Composite attribute → flatten into component attributes.

• Multivalued attribute → separate relation with owner key and multivalued value; usually composite PK.

• Derived attribute → calculate from source attributes when appropriate.

• Recursive relationship → self-referencing FK for a simple 1:N recursive relationship.

• Never infer total participation from cardinality alone; minimum participation must be stated.

Source-specific caveats

The source slides sometimes use simplified schemas with names as keys and sometimes show shorthand cardinality examples. In an exam answer, follow the stated assumptions and clearly label any assumption. The source's explicit tutorial questions are reproduced in meaning in the tutorial section; diagram-only exercises with unreadable or absent requirements are not fabricated as if they were fully legible.